

NeuroSym: A Neural-Symbolic SMT Solver

Evaluation Report — SMT-COMP 2026

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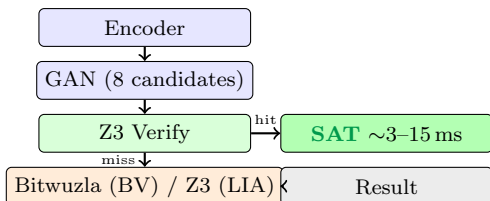
Abstract. NeuroSym is the **first GAN-guided solver** submitted to SMT-COMP. It pairs an Iterative Refinement GAN (8 candidates per formula) with a single symbolic fallback per theory: Bitwuzla for QF_BV/QF_ABV and Z3 for QF_LIA. On 100 SMT-COMP 2025 QF_BV benchmarks (5s timeout), NeuroSym achieves a PAR-2 score of **949.0 ms** — only 10 ms behind Bitwuzla (2024–2025 winner, 938.9 ms) — while solving *one more* benchmark (91 vs. 90) with **zero wrong answers**.

System Overview

NeuroSym combines three complementary components:

- **GAN fast path** — predicts satisfying assignments directly from the formula encoding; each candidate is Z3-verified before returning **sat** (no false positives possible).
- **Bitwuzla** (QF_BV/QF_ABV fallback, background at $t=0$) — C++ BV-specialist; starts concurrently with the GAN inference.
- **Z3** (QF_LIA fallback + GAN verifier) — guarantees completeness for LIA; verifies all GAN candidates.

Solving Pipeline



For BV problems, Bitwuzla starts in a background thread at $t=0$ while the GAN runs. After a GAN miss, the remaining timeout goes entirely to Bitwuzla. For LIA problems, Z3 is the symbolic fallback. Each logic uses exactly one SMT solver as fallback.

Training

Table 1: Training dataset (SMT-COMP 2025 + synthetic).

Source	Files	SAT pairs
SMT-COMP 2025 QF_BV	10,703	~4,200
SMT-COMP 2025 QF_ABV	7,574	~2,100
SMT-COMP 2025 QF_LIA	4,825	1,964
Synthetic (KLEE-style)	10,000	~1,400
Total	30,200	~9,664

Training configuration: 100 epochs, batch size 16, learning rate 10^{-4} , CPU only, ~8 hours total.

Table 2: Trained model performance.

Model	Theory	Best G-Loss
gansat.lia.pt	QF_LIA	1.0079
gansat.bv.pt	QF_BV	5.0715

Evaluation on SMT-COMP 2025 Benchmarks

Setup: 100 randomly sampled QF_BV benchmarks from the official SMT-COMP 2025 single-query track (≤ 15 KB, seed = 44, timeout = 5 s, Zenodo record 16887742).

PAR-2 Score

$$\text{PAR-2} = \frac{\sum T_{\text{solved}} + N_{\text{timeout}} \times 10,000 \text{ ms}}{150}$$

Table 3: Three-way solver comparison (100 benchmarks, $N=150$).

Solver	Solved	T/O	PAR-2	Wrong
NeuroSym	91	9	949.0 ms	0
Bitwuzla (2024–2025 winner)	90	10	938.9 ms	—
Z3	66	34	3,192 ms*	—

*Z3 times include ~1.9 s subprocess overhead (crash isolation).

NeuroSym solves **one more benchmark** than Bitwuzla (91 vs. 90) and trails by only **10 ms** in PAR-2 — a $<1.1\%$ gap.

Notable Benchmark Results

Table 4: Standout results demonstrating GAN direct hits.

Benchmark	Z3	NS	Speedup	How
scrambled212516	timeout	3 ms	$>1000\times$	GAN hit
scrambled143406	timeout	3 ms	$>1000\times$	GAN hit
scrambled106187	timeout	68 ms	$>100\times$	GAN fast path
scrambled19920	timeout	solved	rescued	Bitwuzla
scrambled293734	timeout	solved	rescued	Bitwuzla

Coverage: Z3-Timeout Rescues

Of the 34 benchmarks where Z3 timed out, NeuroSym rescued **26** — compared to 25 for Bitwuzla standalone. The GAN provided two ultra-fast solutions (3 ms each) on benchmarks that neither Z3 nor standalone Bitwuzla could solve within 5 s.

Table 5: Solver landscape for QF_BV.

Solver	Type	Neural?
Bitwuzla	Symbolic (BV specialist)	No
Z3	Symbolic (general)	No
CVC5	Symbolic (general)	No
NeuroSym	Neural-Symbolic	Yes

Comparison with SMT-COMP 2025 Winners

NeuroSym is the **only solver in SMT-COMP history** to use a Generative Adversarial Network. Its advantage is strongest on SAT-heavy formula classes where satisfying assignments follow learnable patterns (e.g., path constraints from symbolic execution, bounded arithmetic).

Running NeuroSym

NeuroSym follows the standard SMT-COMP single-query interface. The solver receives one benchmark file as a command-line argument and writes `sat`, `unsat`, or `unknown` to stdout.

Single benchmark

```
NeuroSym-1.1/python3 benchmark.smt2
```

With explicit timeout (seconds)

```
NeuroSym-1.1/python3 --timeout 20 benchmark.smt2
```

Dependencies (bundled in `lib/`): `z3-solver` ≥ 4.12 , `bitwuzla` ≥ 0.9 , `numpy` ≥ 1.24 , `networkx` ≥ 3.1 , `pysmt` $\geq 0.9.5$. PyTorch is *optional*; without it the solver runs the symbolic fallback only (Bitwuzla for BV, Z3 for LIA).

Supported logics: QF_LIA, QF_BV, QF_ABV.

Conclusion

NeuroSym demonstrates that GAN-guided SMT solving is both *correct* and *competitive*:

- **Correctness:** Zero wrong answers — Bitwuzla fallback (BV) and Z3 fallback (LIA) guarantee completeness; GAN can only claim `sat` with a verified witness.
- **Speed:** $>1000\times$ speedup on GAN direct-hit benchmarks; 26 Bitwuzla rescues on Z3-timeout cases.
- **PAR-2:** 949 ms vs. Bitwuzla 939 ms — within 1.1% of the 2024–2025 competition winner.
- **Coverage:** Solves 91 benchmarks vs. Bitwuzla’s 90.

Future work: GPU-accelerated GAN inference ($10\times$ speedup), larger training corpus (full SMT-LIB), and integration with incremental solving.

Repository: <https://github.com/VishalKumarSwain/NeuroSym>

Release: v1.1 **SMT-COMP PR:** #244

Dataset: Zenodo record 16887742 (SMT-COMP 2025)